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Evaluating clinical competencies of large language models with a general practice benchmark

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Large Language Models (LLMs) have demonstrated considerable potential in general practice. However, existing benchmarks and evaluation frameworks primarily depend on exam-style or simplified question-answer formats, lacking a competency-based structure aligned with the real-world clinical responsibilities encountered in general practice. Consequently, the extent to which LLMs can reliably fulfill the duties of general practitioners (GPs) remains uncertain. In this work, we propose a novel evaluation framework to assess the capability of LLMs to function as GPs. Based on this framework, we introduce a general practice benchmark (GPBench), whose data are meticulously annotated by domain experts in accordance with routine clinical practice standards. We evaluate ten state-of-the-art LLMs and analyze their competencies. Our findings indicate that current LLMs are not suitable for autonomous deployment in clinical general practice and that all realistic applications require continuous human oversight; further optimization specifically tailored to the daily responsibilities of GPs remains essential.

Large Language Models (LLMs) have emerged as a prominent technology in recent years, demonstrating remarkable advancements in mathematical reasoning, drug discovery, chemical property prediction, and various other domains. Owing to their extensive knowledge bases and sophisticated reasoning capabilities, LLMs hold substantial potential for applications in the healthcare sector[1,2]. MedPaLM[3] and MedPaLM 2[4] have reported performance on specific medical tasks that rivals or even surpasses that of human experts. GPT-4 outperformed medical students in open-ended clinical reasoning examinations[5]. ChatGPT demonstrated a performance level comparable to that of a third-year medical student on evaluations of core medical knowledge[6]. Med-PaLM[3] was the first model to attain a passing score on the MedQA dataset[7], which consists of questions modeled after the U.S. Medical Licensing Examination (USMLE). AMIE[8] has also demonstrated effectiveness in assisting physicians with generating potential disease diagnoses.

To evaluate whether LLMs can serve as general practitioners (GPs), it is essential to consider the distinct characteristics of this clinical domain. General practitioners follow a unique framework of clinical thinking: it emphasizes symptom-oriented differential diagnosis rather than definitive diagnosis, considers the interplay among multiple physiological systems rather than focusing solely on organ-specific pathologies, and requires a holistic application of the biopsychosocial model. Moreover, given the context in which LLMs serve as GPs, their responses should include accurate disease identification and comprehensive diagnoses, offer personalized treatments with clearly defined targets, and incorporate relevant health management considerations.

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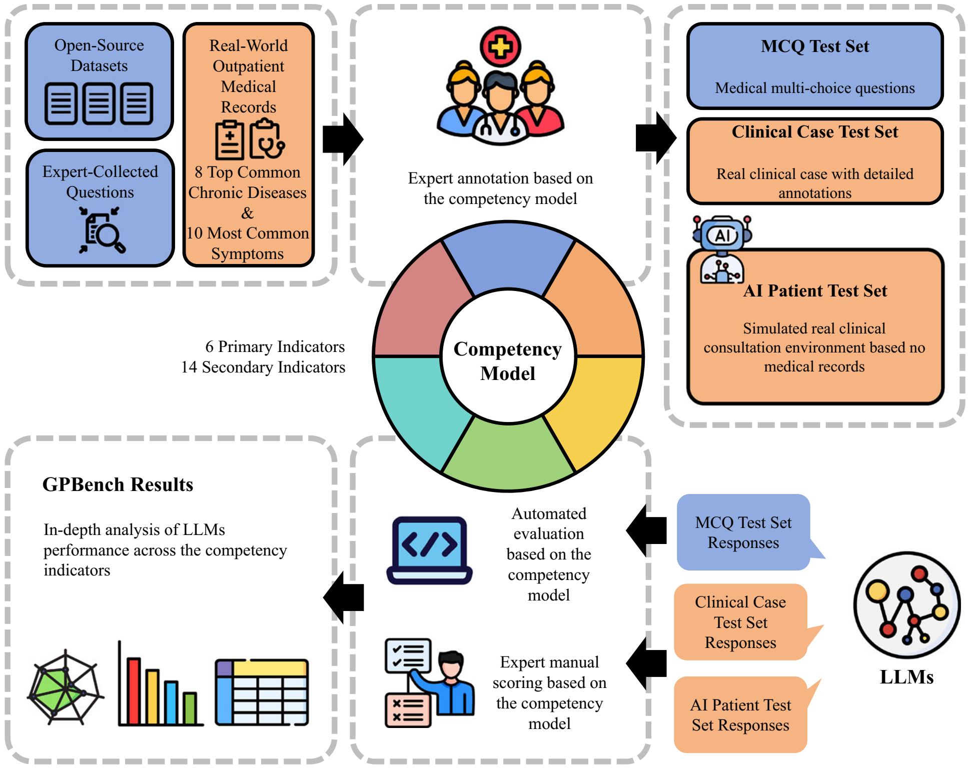

Fig. 1 | An overview of the general practice benchmark (GPBench). GPBench comprises a competency model and an evaluation dataset. The competency model delineates six primary indicators and fourteen secondary indicators, capturing the core competencies essential for routine general practice. Based on the competency model, we collected data from open-source datasets and real outpatient medical records from Tertiary A-grade hospitals to create an evaluation dataset, which contains three parts: the Multiple-choice Question (MCQ) Test Set, Clinical Case Test Set, and Artificial Intelligence (AI) Patient Test Set. Ground truth and scoring criteria for each case were annotated in detail by experts. Accuracy is used as the evaluation metric for the MCQ Test Set, while for the other two test sets, a three-member general practitioner panel grades responses based on the annotated scoring criteria.

Most existing benchmarks for LLMs still rely on exam-style formats[9–11]. Although recent studies have introduced more clinically oriented evaluations[12,13], these efforts remain largely confined to specialist-level tasks and thus fail to reflect the open-ended reasoning and decision-making integral to real-world general practice, where physicians must ensure continuity of care, triage under uncertainty, and manage multimorbidity in community settings.

To address the aforementioned issues, we propose a general practice benchmark (GPBench) to evaluate LLMs in a manner analogous to the evaluation of GPs. As illustrated in Fig. 1, GPBench comprises a competency model and an evaluation dataset. The competency model, detailed in Fig. 2, was derived from real-world clinical practice and delineates the core competencies and decision-making processes essential to routine general practice. It defines six primary indicators—basic medical knowledge, diagnosis, decision-making, health management, health economics, and medical ethics and humanities—and fourteen secondary indicators that specify measurable capabilities under each primary category. The evaluation dataset consists of three components: a Multiple-choice Question (MCQ) Test Set, a Clinical Case Test Set, and an Artificial Intelligence (AI) Patient Test Set. Additionally, we constructed a separate development set, which remained completely independent from the evaluation dataset and was excluded from all testing procedures. All evaluation tasks in this study were conducted in Chinese to ensure consistency with the working language of both physicians and annotators.

Results

We used GPBench to evaluate representative models, as shown in Table 1, including general-purpose models, reasoning models, and medical-specialist models. A reasoning model refers to a type of LLM specifically designed or trained to emulate human-like reasoning processes, thereby exhibiting enhanced reasoning capabilities. A medical-specialist model was an LLM fine-tuned using carefully curated data from the medical domain. The evaluation experiments were conducted using the default hyperparameters available on the official release websites or those recommended by the respective providers (see Supplementary Table 1 for details). We also conducted a human evaluation involving general practitioners to provide a reference for interpreting the performance of LLMs.

Results on the three test sets

The results of reasoning models, general models, and medical specialists on GPBench are discussed based on the indicators defined in the constructed competency model. To enable a more fine-grained capability analysis, some analyses are further conducted at the level of secondary indicators. An overview of all indicators in GPBench is provided in Fig. 2 (see Supplementary Data 1 for detailed descriptions).

The evaluation on the MCQ Test Set encompassed 6 primary indicators and 14 secondary indicators. The performance of various LLMs with respect to the primary indicators is shown in Fig. 3, where the results for each primary indicator were calculated as the weighted average of the corresponding secondary indicators (see Supplementary Table 7 for more details). Overall, reasoning models, DeepSeek-R1 (82.74 ± 1.20) and o1-preview (79.16 ± 1.27), demonstrated relatively strong performance, albeit with certain limitations. Specifically, DeepSeek-R1 exhibited balanced performance across indicators but did not reach a satisfactory level. In contrast, o1-preview, with accuracy falling below 80 in several indicators such as I3, I5, and I6, showed varying adaptability to different tasks. This suggested that although

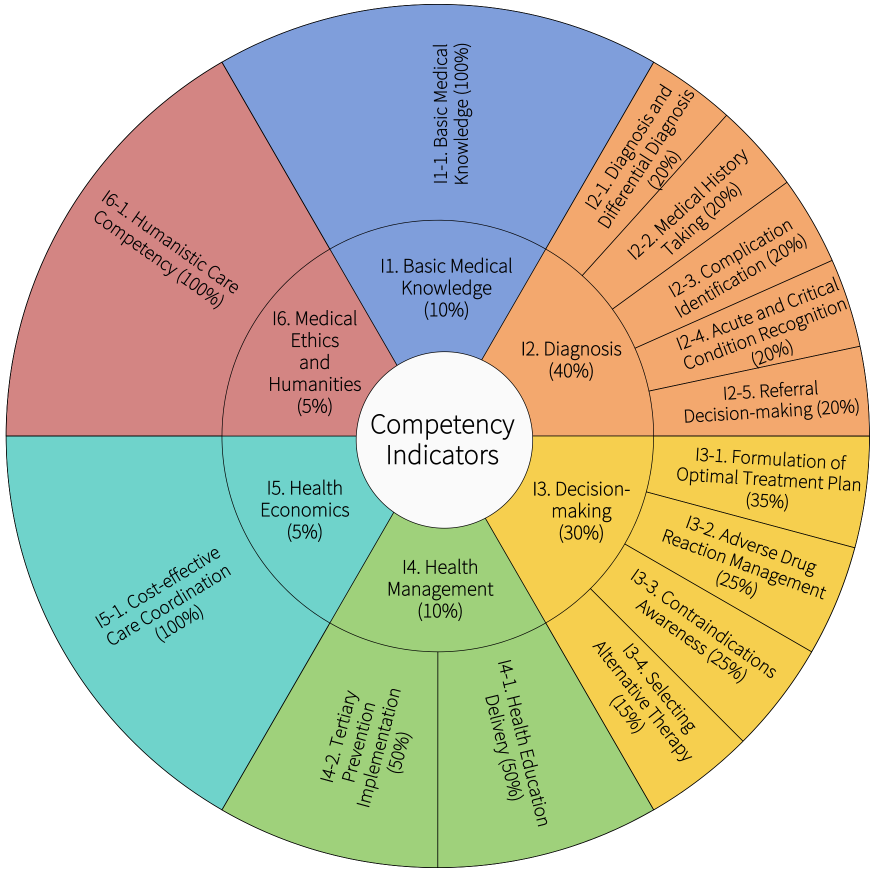

Fig. 2 | An overview of the competency indicators and their associated importance weights in our evaluation framework.

Table 1 | The information of large language models (LLMs) evaluated in our experiments

Model	Parameter	Chinese Corpus Proportion	Supported Languages	Type	
				Medical Specialist	Reasoning Model
GPT-4o[30]	–	Not Reported	Multilingual	No	No
GPT-4-turbo	–	Not Reported	Multilingual	No	No
o1-preview[31]	–	Not Reported	Multilingual	No	Yes
Gemini-1.5-pro[32]	–	Not Reported	Multilingual	No	No
Qwen2.5-7B-Instruct[33]	7B	Not Reported	Multilingual	No	No
Qwen2.5-72B-Instruct[33]	72B	Not Reported	Multilingual	No	No
Claude-3.5-Sonnet	–	Not Reported	Multilingual	No	No
DeepSeek-V3[34]	671B	Not Reported	Multilingual	No	No
DeepSeek-R1[35]	671B	Not Reported	Multilingual	No	Yes
HuatuoGPT-o1-7B[36]	7B	Not Reported	Chinese/English	Yes	Yes

"Multilingual" indicates models capable of understanding multiple languages, including Chinese.

models with advanced reasoning capabilities exhibited greater potential, they still faced challenges in fine-grained decision-making tasks. In addition, the overall performance of general-purpose LLMs was relatively weaker than that of reasoning-optimized LLMs, as evidenced by the comparison between their respective top-performing models (Qwen2.5-72B-Instruct: 78.13 ± 1.28 vs. DeepSeek-R1: 82.74 ± 1.20; Welch's t-test, p = 0.0043). Meanwhile, Qwen2.5-72B-Instruct significantly outperformed Qwen2.5-7B-Instruct (78.13 ± 1.28 vs. 69.63 ± 1.40; Welch's t-test, p < 0.0001), indicating that LLMs with larger parameter scales tended to achieve better performance.

The medical specialist HuatuoGPT-o1-7B demonstrated strong performance in I4 (80.23 ± 3.81), I5 (81.82 ± 8.22), and I6 (86.76 ± 3.70), whereas its results in the remaining three primary indicators—I1 (70.59 ± 1.10), I2 (72.85 ± 2.30), and I3 (70.45 ± 2.58)—were at a moderate level. This disparity suggested that current medical specialist LLMs were fine-tuned without comprehensive optimization across diverse medical competencies.

For the Clinical Case Test Set, the inputs of LLMs were medical records, and the target outputs were the corresponding diagnosis and treatment. This test set was specifically designed to reflect the real-world demands of general practice. Representative cases were selected for inclusion, with the aim of covering as many different diseases, symptoms, and levels of difficulty as possible. As a result, each case exhibited significant attribute variation, which in turn affects the performance of LLMs, leading to considerable fluctuation in test results across different cases. On this test set, some secondary indicators (e.g.,

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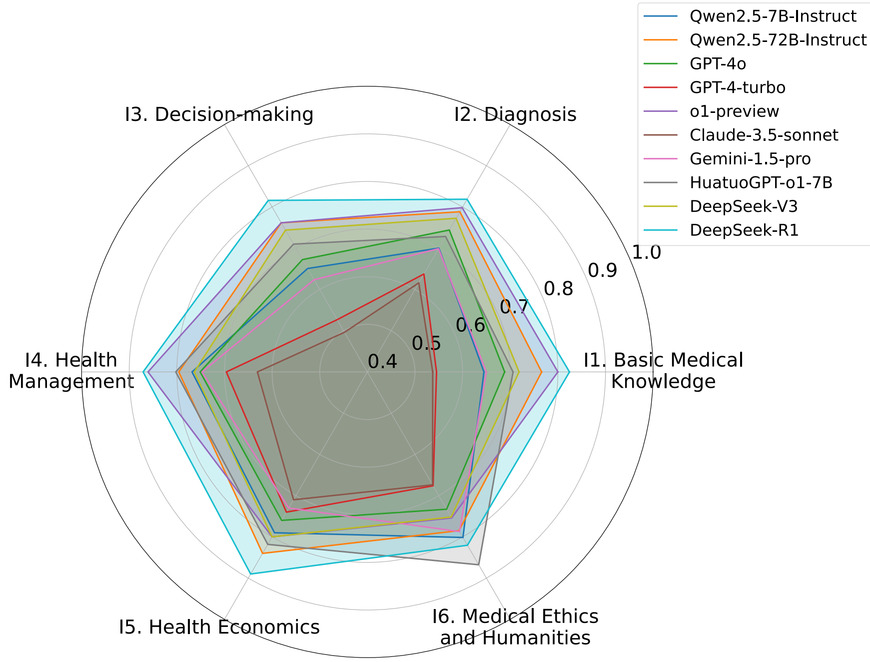

Fig. 3 | The performance of large language models (LLMs) on the Multiple-choice Question (MCQ) Test Set across the primary competency indicators.

I1–1, I2–2, I4–2, I5–1, I6–1) could not be evaluated. Thus, six secondary indicators—those that were manifested in the generations—were selected for the final evaluation. Figure 4 presents the performance of LLMs across multiple competency indicators (see Supplementary Table 8 for more details).

We found that most LLMs showed significant shortcomings in the core aspects of competencies. Specifically, in the I2-1 indicator, 7/10 models scored below 70, indicating that most LLMs failed to construct a complete chain of diagnostic reasoning. In the I3-1 indicator, only DeepSeek-R1 (77.04 ± 21.33) scored above 70, indicating that current LLMs were basically unable to generate satisfactory treatment plans.

DeepSeek-R1 outperformed other models across indicators related to diagnosis and treatment, ranking at the forefront across all indicators (DeepSeek-R1: 81.80 ± 16.92 vs. Gemini-1.5-pro: 76.46 ± 20.14; Wilcoxon rank-sum test, $p = 0.046$), with Gemini-1.5-pro demonstrating the second-best performance. This statistically significant difference highlighted a potential direction for optimizing LLMs in general practice applications. Furthermore, the results underscored the superiority of reasoning-optimized LLMs over general-purpose LLMs, as DeepSeek-R1 and Gemini-1.5-pro represented the top-performing models within their respective model categories. Meanwhile, Qwen2.5-72B-Instruct outperformed Qwen2.5-7B-Instruct (72.79 ± 17.18 vs. 68.63 ± 16.26; Wilcoxon rank-sum test, $p = 0.010$), suggesting that larger models tended to achieve better performance.

Moreover, despite explicit prompts, none of the LLMs proactively demonstrated I5 or I6. This suggested that current LLMs remained limited to a simplified framework of diagnosis and treatment, lacking a broader understanding of the social attributes of healthcare services. We further revealed the deficiencies of LLMs through a detailed analysis of the responses, as described in section 2.3.

The AI Patient Test Set evaluated LLMs' competency in medical history taking by measuring their ability to ask relevant questions. Figure 5 shows the resulting competency scores. DeepSeek-R1 was excluded due to its substantial failure to follow instructions and meet basic evaluation criteria.

From Fig. 5, we found that in the I2-2 indicator, all models scored below 60, with the best performer, o1-preview, achieving only 55.20 ± 8.01. LLMs exhibited significant deficiencies in medical history taking, ultimately hindering their ability to support effective clinical decision-making. In one of the abdominal pain tests, o1-preview showed significant weaknesses. It failed to adequately explore critical pain features such as intensity, radiation, and triggering or relieving factors—essential for distinguishing conditions like appendicitis and cholecystitis. The assessment of associated symptoms was superficial, limited to checking for vomiting and fever, without further inquiry into important diagnostic clues like the nature of the vomitus or the presence of chills. Additional issues included the omission of other potentially relevant symptoms, incomplete collection of personal and medication history, and failure to assess genitourinary, respiratory, and gastrointestinal systems. Past medical and family histories were also insufficiently gathered.

Human performance as a reference

To benchmark the performance of LLMs against human physicians, we evaluated GPs using the same test sets. The participating GPs were classified into three groups: Group A (>10 years), Group B (5 ~ 10 years), and Group C (<5 years). This stratification allowed us to position the capabilities of LLMs within a meaningful spectrum of real-world clinical expertise. The scores of the human evaluation can be found in Supplementary Table 9.

On the MCQ Test Set, LLMs generally outperformed human participants in terms of accuracy. This outcome was anticipated, as the format restricts possible answers and mainly tests factual knowledge recall, an area where LLMs excel owing to their extensive pretraining on large-scale factual text corpora.

In contrast, on the Clinical Case Test Set, LLMs performed worse than human physicians overall. Among the evaluated models, the best-performing LLM, DeepSeek-R1, reached a competency level comparable to that of Group C human physicians, while the remaining models exhibited inferior performance. Competency analysis revealed that LLMs had substantial limitations in core diagnostic and treatment decision-making capabilities, falling short even when compared to Group C human physicians. Nonetheless, LLMs demonstrated a distinct advantage in health education, outperforming experienced Group A human physicians—likely attributable to their ability to organize and present information in a well-structured manner.

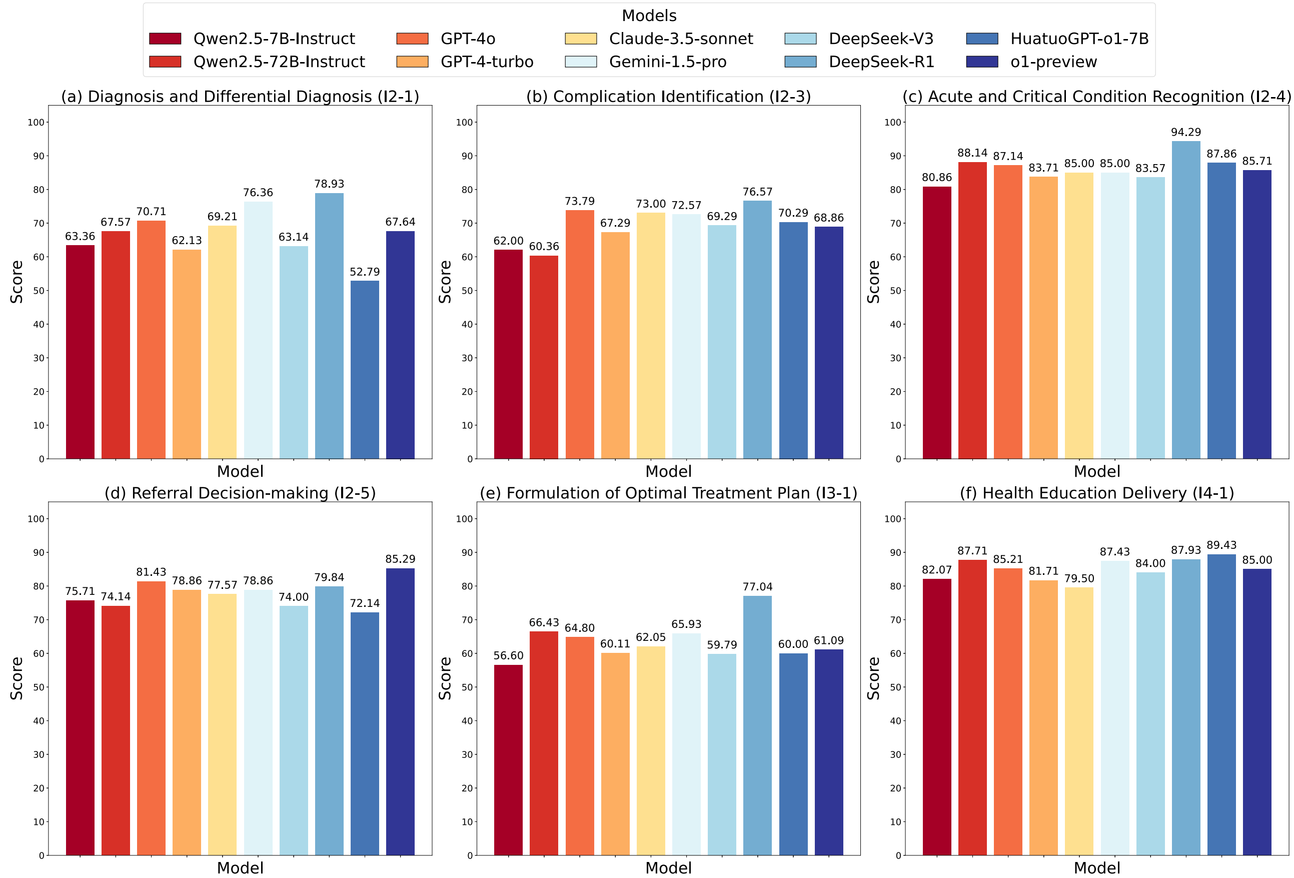

Fig. 4 | The performance of large language models (LLMs) on the Clinical Case Test Set across multiple competency indicators. **a** Performance on diagnosis and differential diagnosis (I2-1). **b** Performance on complication identification (I2-3). **c** Performance on acute and critical condition recognition (I2-4). **d** Performance on referral decision-making (I2-5). **e** Performance on formulation of optimal treatment plan (I3-1). **f** Performance on health education delivery (I4-1).

The gap further widened in the medical history taking competency on the AI Patient Test Set, where even the highest-performing model, o1-preview, underperformed compared to Group C human physicians. This limitation likely stems from LLMs' lack of real clinical experience and insufficient exposure to temporally interactive diagnostic processes, leading to logical discontinuities and incomplete or fragmented histories.

Overall, these results indicate that while LLMs have matched or even surpassed human physicians in assessments of static knowledge, they still lag significantly behind in interactive and reasoning-intensive clinical tasks. Their strengths lie in factual recall and the clear articulation of information, yet they have not attained the practical competency level of even less experienced general practitioners in real-world diagnostic and patient-interview scenarios.

Competency analysis

We analyzed LLM responses on the Clinical Cases Test Set using the proposed evaluation framework to identify key deficiencies in core competencies. For diagnosis and treatment recommendations, we reviewed all outputs and categorized common deficiency types. Each case was annotated with the relevant deficiencies per LLM, allowing for multiple deficiencies per case. For each deficiency type, we computed the proportion of errors per LLM across applicable cases. Results are shown in Tables 2, 3, respectively.

We conducted a detailed examination of LLM-generated diagnostic outputs and identified common deficiencies in diagnostic competency. Our analysis was presented as follows.

(1) Lack of accurate disease grading, staging, and risk stratification. LLMs tended to oversimplify the management of diseases that required classification or risk stratification. As shown in Table 2, such a deficiency was common in LLMs with Qwen2.5-7B-Instruct showing the highest proportion at 20/27. In the case tests for hypertension (HTN), coronary artery disease (CAD), and chronic kidney disease (CKD), most LLMs, including Qwen2.5-72B-Instruct, o1-preview, and DeepSeek-R1, failed to perform grading, staging, and risk stratification according to the corresponding specialty guidelines. Specifically, in one of the CAD case tests, 7/10 models only provided a general diagnosis of "coronary artery disease/stable angina" without specifying cardiac function classification. Moreover, some models even exhibited incorrect disease grading and staging. For example, in one of the CKD case tests, Qwen2.5-72B-Instruct and GPT-4o misdiagnosed CKD stage 5 as CKD stage 3.

(2) Hallucination on disease grading, staging, and risk stratification. LLMs arbitrarily graded and staged diseases without clinical guidelines or evidence-based medical support, leading to a significant issue of fabricating clinical staging during diagnosis, which involved a variety of diseases such as pneumococcal pneumonia, diabetes, gouty arthritis, osteoarthritis, pneumothorax, and acute appendicitis. For example, in one of the abdominal pain case tests, Qwen2.5-72B-Instruct provided the diagnoses of "acute appendicitis Grade 0". As shown in Table 2, Qwen2.5-72B-Instruct exhibited the highest proportion 42/43, with DeepSeek-R1 also reaching 18/43. In contrast, HuatuoGPT-o1-7B demonstrated a much lower proportion of only 1/43, suggesting that LLMs that had undergone domain-specific fine-tuning were considerably less susceptible to hallucinations.

(3) Blind spots in the identification of comorbidities and complications. Current LLMs exhibited insufficient awareness in systematically screening for comorbidities and complications, especially in the scenarios of metabolic and cardiovascular diseases, where key diagnostic elements were frequently omitted, with a deficiency proportion approximately approaching 50% among applicable cases involving this deficiency (N_3 = 45) for LLMs shown in Table 2. This issue was mainly manifested in two aspects: Failed to fully identify comorbidities and complications. For example, in one of the hyperlipidemia (HLD) case tests, almost all models failed to diagnose additional comorbidities or complications, with the absence of a diagnosis of severe fatty liver and HLD. Incorrectly assess comorbidities and complications. For instance, in one of the HLD case tests, Qwen2.5-72B-Instruct and Gemini-1.5-pro incorrectly diagnosed diabetic nephropathy (DN), although the patient did not meet the diagnostic criteria for DN.

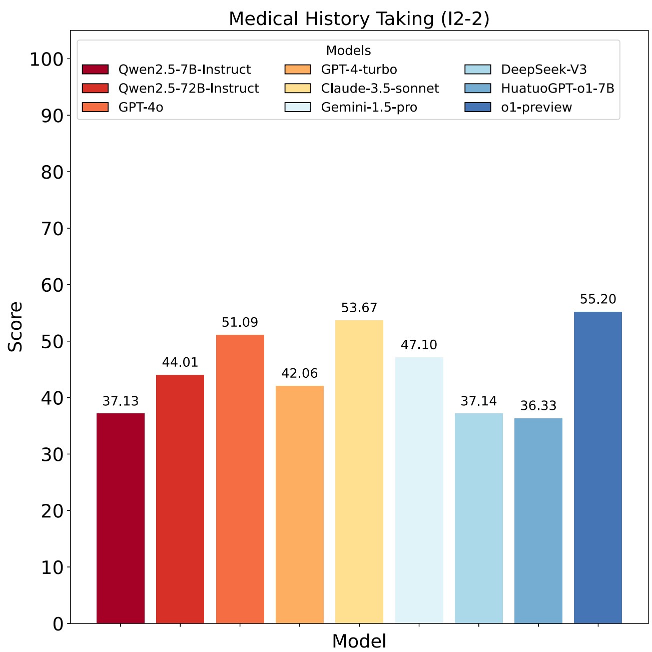

Fig. 5 | The performance of large language models (LLMs) on the Artificial Intelligence (AI) Patient Test Set for the medical history taking (I2-2) indicator.

(4) Deficiency in acute and critical condition assessment. LLMs showed significant shortcomings in the identification of clinical emergencies, particularly in the recognition of time-sensitive acute and life-threatening critical conditions, with a deficiency proportion of approximately 10% among the applicable cases (N_4 = 18) according to Table 2.

(5) Lack of accurate diagnosis for rare diseases. LLMs exhibited limitations in diagnosing rare diseases in our evaluations, as rare diseases were often associated with complex pathological mechanisms and heterogeneous clinical manifestations. For instance, in one of the fever case tests, 8/10 LLMs failed to provide an accurate diagnosis for the rare disease scrub typhus.

We examined all the details of outputs generated by the LLMs and identified common deficiencies in treatment decision-making, with the results shown in Table 3. Our analysis was presented as follows.

(1) Lack of treatment goals. In the treatment recommendations generated by LLMs for chronic diseases (e.g., HTN and diabetes mellitus (DM)), there was a widespread lack of clearly defined therapeutic targets. As shown in Table 3, this deficiency type accounted for approximately half of the applicable cases (N_6 = 28). For instance, nearly all models failed to explicitly specify control targets for blood pressure or blood glucose in their treatment plans.

(2) Pharmacotherapy risks. Through case analysis, we found that LLMs exhibited issues related to pharmacotherapy risks, with approximately 40% of the applicable cases (N_7 = 70) demonstrating this deficiency type according to Table 3. We described four types of such deficiencies as follows.

- Omission of core drugs. This was the most common type of pharmacotherapy risk. Most LLMs omitted the main drugs in the specific disease recommendations, such as CAD, HLD, and CKD. For example, in one of the HLD case tests, 9/10 models failed to

Table 2 | The proportion of observed deficiencies of large language models (LLMs) in the diagnostic process on the Clinical Case Test Set

Model	Lack of Accurate Disease Grading, Staging, and Risk Stratification (Applicable cases N_1 = 27)	Hallucination on Disease Grading, Staging, and Risk Stratification (Applicable cases N_2 = 43)	Blind Spots in Identification of Comorbidities and Complications (Applicable cases N_3 = 45)	Deficiency in Acute and Critical Condition Assessment (Applicable cases N_4 = 18)	Lack of Accurate Diagnosis for Rare Diseases (Applicable cases N_5 = 3)
Qwen2.5-7B-Instruct	20/27	16/43	23/45	3/18	2/3
Qwen2.5-72B-Instruct	14/27	42/43	13/45	1/18	1/3
GPT-4o	9/27	11/43	26/45	2/18	2/3
GPT-4-turbo	13/27	6/43	25/45	3/18	3/3
Claude-3.5-Sonnet	11/27	17/43	26/45	3/18	2/3
Gemini-1.5-pro	11/27	14/43	19/45	2/18	1/3
DeepSeek-V3	15/27	15/43	22/45	3/18	2/3
DeepSeek-R1	7/27	18/43	8/45	4/18	2/3
HuatuoGPT-o1-7B	12/27	1/43	31/45	2/18	2/3
o1-preview	15/27	18/43	17/45	2/18	1/3

Table 3 | The proportion of observed deficiencies of large language models (LLMs) in treatment recommendations on the Clinical Case Test Set

Models	Lack of Treatment Goals (Applicable cases N_6 = 28)	Pharmacotherapy Risks (Applicable cases N_7 = 70)				Blind Spots in Non-Pharmacological Interventions (Applicable cases N_8 = 70)	Insufficient Standardization in Critical Illness Management (Applicable cases N_9 = 18)
		Omission of Core Drugs	Neglect of Drug Interaction Contraindications	Lack of Drug Dosage Guidance	Improper control of medication indications		
Qwen2.5-7B-Instruct	17/28	17/70	1/70	12/70	6/70	14/70	9/18
Qwen2.5-72B-Instruct	16/28	18/70	1/70	7/70	3/70	9/70	8/18
GPT-4o	14/28	18/70	4/70	6/70	2/70	11/70	8/18
GPT-4-turbo	17/28	21/70	2/70	17/70	2/70	11/70	9/18
Claude-3.5-Sonnet	18/28	24/70	2/70	12/70	2/70	10/70	8/18
Gemini-1.5-pro	11/28	16/70	1/70	10/70	0/70	11/70	7/18
DeepSeek-V3	18/28	20/70	5/70	5/70	1/70	12/70	8/18
DeepSeek-R1	12/28	5/70	2/70	2/70	1/70	12/70	2/18
HuatuoGPT-o1-7B	17/28	20/70	0/70	10/70	3/70	10/70	6/18
o1-preview	15/28	13/70	2/70	4/70	0/70	10/70	5/18

include somatostatin and proton pump inhibitors (PPIs) in the treatment plan for severe acute pancreatitis.

- Neglect of drug interaction contraindications. Some LLMs failed to comprehensively evaluate the overall safety of drug combinations, ignoring the potential for increased side effects or adverse reactions. For example, in one of the HTN case tests, Qwen2.5-7B-Instruct recommended the concurrent use of fibrate and statin drugs (fenofibrate and atorvastatin) for lipid-lowering.
- Lack of drug dosage guidance. Some LLMs showed insufficient or erroneous guidance on drug dosing. For example, in one of the HLD case tests, GPT-4-turbo failed to specify the dosages for lipid-lowering drugs and antibiotics. In the same case, Qwen2.5-72B-Instruct recommended a dose of Ulinastatin at 10,000 U, which significantly deviated from the 100,000 U recommended by the guidelines. Moreover, in some cases, certain models even suggested dosing routes that did not match the acute phase of the disease.
- Improper control of medication indications. Some LLMs failed to adequately consider individual patient characteristics and specific disease requirements when providing treatment recommendations. For example, in one of the CKD case tests, LLMs, including Qwen2.5-7B-Instrcut, Qwen2.5-72B-Instrcut, and Claude-3.5-Sonnet, did not account for the patient's renal function when selecting and adjusting antihyperglycemic drugs.

(3) Blind spots in non-pharmacological interventions. We found that the treatment recommendations of LLMs for non-pharmacological treatment plans, such as interventional therapies and surgical indications, were significantly under-represented. As shown in Table 3, the deficiency type accounted for approximately 15% among the applicable cases (N_8 = 70). For instance, in the case tests for CAD and CKD, LLMs, such as Qwen2.5-72B-Instruct, GPT-4o, and Gemini-1.5-pro, did not provide interventional therapies, dialysis plans, and other options tailored to the disease progression stage of patients.

(4) Insufficient standardization in critical illness management. In the management recommendations for acute and critical conditions, LLMs failed to adhere to clinical guidelines and even omitted critical steps in certain disease scenarios, with the highest proportion of 9/18 for Qwen2.5-7B-Instruct and GPT-4-turbo.

We examined the model-generated health management advice and found that current LLMs tended to adopt an overly principled approach when providing health education. Although the content generally met basic scoring standards, it often remained at a broad, principle-based level, lacking concrete, actionable guidance that would effectively support clinical practice needs. For instance, in the case tests for HTN, most LLMs merely suggested a "sodium-restricted diet" without specifying precise daily sodium intake limits (e.g., ≤ 5*g*).

We examined the model-generated outputs and identified the presence of medical ethical risks, which were primarily manifested as an over-treatment tendency and an increased risk of misdiagnosis or missed diagnosis.

(1) Over-treatment tendency. Some LLMs tended to generate unreasonable treatment recommendations due to algorithmic bias or limitations in training data. This was particularly evident in specific disease scenarios. For example, in one of the HTN case tests, Qwen2.5-7B-Instruct recommended combination therapy with two antihypertensive drugs and two lipid-lowering medications, which was unnecessary for the current patient and inconsistent with clinical guidelines.

(2) Risk of misdiagnosis or missed diagnosis. In one of the HTN case tests, the patient was experiencing a hypertensive emergency, but some models misdiagnosed the condition as mild hypertension or an unrelated disorder, such as Qwen2.5-7B-Instruct diagnoses hypertension, GPT-4-turbo diagnoses transient ischemic attack (TIA).

Discussion

The evaluation revealed that LLMs generally outperformed human physicians on knowledge-intensive tasks but underperformed in reasoning and interactive competencies critical to general practice. Specifically, LLMs faced challenges in diagnosing, managing multimorbidity, and making treatment decisions, including overlooking key therapeutic objectives, deviating from standard protocols, and failing to construct coherent reasoning frameworks during history-taking, which also raised ethical concerns such as overtreatment or misdiagnosis. In contrast, human physicians showed gradual improvement with experience, with experienced general practitioners demonstrating more coherent reasoning and reliable clinical judgment.

Potential causes of the deficiencies identified in our study include the following aspects. (1) The hallucination problem inherent in large models. Hallucinations refer to outputs that appear plausible or logical but are, in fact, incorrect, fabricated, or inconsistent with established facts. This issue primarily stems from the fact that LLMs generate responses based on linguistic statistics rather than knowledge-based logic. (2) The lack of modeling of human physiology, disease progression, and pharmacology. LLMs are trained via next-token prediction over large textual corpora, and therefore lack true capabilities in modeling human physiological and pathophysiological processes. As a result, they often struggle to accurately infer disease trajectories or assess comorbid conditions (e.g., drug interactions in polypharmacy). (3) Deficiencies in fine-grained information retention and reasoning. This limitation likely stems from the scarcity of structured, case-level data derived from human clinical training and workflows in the models' training corpora.

Based on the preceding analysis, we propose the following recommendations for the application of LLMs in general practice:

- LLMs should only be used under the supervision of medical personnel capable of evaluating the correctness of the output.
- Integrate local clinical guidelines, case studies, and textbooks to support evidence-based medicine, thereby reducing hallucinations and improving factual accuracy, potentially combined with automated fact-checking mechanisms[14,15].
- Prefer LLMs fine-tuned for medical reasoning, with sufficient parameter scale, and consider interactive or multi-agent architectures for complex decision-making[16,17].
- For languages with limited pretraining data, fine-tune LLMs on high-quality local corpora to enhance reliability and reasoning performance.

We have shown the limitations of LLMs in real-world general practitioner scenarios and now discuss the main limitations of our study. First, all cases and prompts were in Chinese and collected in China, which may introduce language-related biases. The models' performance in other languages and regions requires further validation. Second, we used standard prompting, the most common approach for applying LLMs. Performance may improve with advanced methods, such as interactive multi-agent systems[18] and reinforcement learning tailored to medicine[12], though these are beyond the scope of this study, which focuses on evaluating LLMs' native medical capabilities. Lastly, our dataset includes samples covering 8 diseases and 10 symptoms, as all samples require fine-grained human annotation. To better reflect the nature of GPs' daily tasks, we plan to develop automated scoring methods to reduce reliance on expert annotation and expand disease coverage.

In conclusion, by leveraging the proposed evaluation method and our fine-grained annotated dataset, we evaluated the performance of current state-of-the-art large models, encompassing both general-purpose and medical-specialized variants. The experimental results indicate that the models assessed in this study still exhibit significant limitations in critical tasks such as clinical decision support, diagnostic accuracy, and treatment recommendation. To improve the performance of large models in real-world general practice scenarios, we propose that future research should focus on the following directions:

- Further enhancing the systematic medical knowledge (i.e., diseases, drugs, symptoms) and complex reasoning capabilities of large models, while mitigating the occurrence of medical hallucinations in LLMs[19].
- Progressively learn from physicians' clinical experience by incorporating structured data derived from clinical workflows into the training process.
- Strengthening capabilities in human body modeling, disease progression modeling, and pharmacological modeling[20].

Our GPBench can effectively reflect the limitations of applying LLMs in general practice and provide valuable guidance for enhancing their capabilities. The dataset will be publicly released to support broad adoption. Through this initiative, we aim to promote the development of LLMs for general practice and contribute to the advancement and optimization of clinical decision support systems.

Methods

Ethics approval and consent to participate

This study was approved by the Institutional Review Boards of The Sixth Affiliated Hospital of Sun Yat-sen University (IRB No.2025ZSLYEC-117) and Xinyi People's Hospital (IRB No.CT-2025-03-01). The requirement for informed consent was waived due to the anonymized nature of the data and minimal risk to participants. All patient identifiers were removed prior to analysis. Data governance and anonymization procedures were conducted under the oversight of the participating hospitals' institutional data protection and ethics governance frameworks, in compliance with the General Data Protection Regulation (GDPR). The authors confirm no conflicts of interest related to this work.

Evaluation framework

To evaluate LLMs in a manner analogous to the assessment of GPs, we draw upon widely accepted competency frameworks in general practice, including WONCA[21], ACGME[22], the Iceberg Model[23], and the Onion Model[24], and established an evaluation framework using the Delphi expert consensus method. We formed an expert committee composed of ten general medicine experts and two medical informatics experts (see Supplementary Table 2). The committee conducted a comprehensive analysis of the core competencies required for GPs in daily practice and selected indicators that possess both technical characteristics and clinical applicability, grounded in the practical deployment of LLMs in general practice. Competencies in non-technical domains, such as organizational management, leadership, and teamwork, were excluded. After three rounds of iterative consultation, the committee proposed a competency model specifically tailored for LLM evaluation. The resulting competency model for LLMs comprised six primary general practice competency indicators and fourteen secondary indicators, each associated with a corresponding importance weight. The complete competency model is illustrated in Fig. 2 (see Supplementary Data 1 for a detailed description of each indicator). It encapsulated the core competencies essential for routine clinical activities of GPs, which also aligned with the expected capabilities of LLMs in real-world scenarios.

Test set construction

Based on the specially designed evaluation framework, we constructed data sets to reflect the competency of LLMs across various dimensions. An overview of the dataset is presented in Table 4. All experiments were conducted in Chinese, including the medical records, prompts for LLMs, model responses, scoring criteria, and scoring procedures. This design aligned with the clinical environment and the working language of Chinese medical professionals. All models were evaluated using default inference parameters and an identical prompt design, which adhered to widely accepted prompt engineering practices. The

Table 4 | Overview of the proposed benchmark

Test Set	Content	Format	Number
MCQ Test Set	Open-source data and expert supplementation	Objective questions	3661
Clinical Case Test Set	Outpatient medical records containing any of the following 8 major chronic diseases or 10 common symptoms: • 8 Major Chronic Diseases: –Hypertension (HTN) –Hyperlipidemia (HLD) –Coronary Artery Disease (CAD) – Chronic Kidney Disease (CKD) – Chronic Obstructive Pulmonary Disease (COPD) – Cerebrovascular Disease (CVD) – Diabetes Mellitus (DM) –Cancer (CA)	Open-ended generation	70
AI Patient Test Set	• 10 Common Symptoms: – Fever – Edema – Emaciation – Chest Pain – Headache – Abdominal Pain – Hematochezia – Joint Pain – Jaundice – Cough	Open-ended interaction	70

prompt template incorporated "role definition, possessed skills, and task objectives." The prompts were tuned on a separate development set to ensure the validity of the evaluation on the test sets (see Supplementary Method 1, Supplementary Method 2). The proposed GPBench evaluation dataset consisted of three components: the Multiple-choice Questions (MCQ) Test Set, the Clinical Case Test Set, and the AI Patient Test Set, each of which had a corresponding development set (see Supplementary Method 3). The following describes the construction of the test sets.

The MCQ Test Set comprised 3661 multiple-choice questions designed to evaluate LLMs' foundational medical knowledge and theoretical understanding. The Clinical Case Test Set, drawn from 70 real outpatient records, used open-ended case analysis to assess LLMs' ability to develop systematic diagnostic and treatment plans, highlighting their handling of complex clinical problems and potential blind spots. The AI Patient Test Set, also based on the same 70 cases, simulated interactive consultations with AI-generated "patients" to evaluate LLMs' clinical responsiveness and decision-making in realistic outpatient scenarios.

The MCQ test set was constructed by selecting general practice-related questions from CMB[25] and MedBench[26], and annotating them with secondary indicators from the proposed competency model (see Supplementary Table 3). General practice experts manually supplemented them with questions from additional sources to address the problem that some indicators had few associated questions. The final distribution is shown in Fig. 6.

The Clinical Case Test Set consisted of 70 real-world outpatient records from two medical centers. Based on epidemiological data[27–29], eight common chronic diseases and ten frequent clinical symptoms were selected, with 3–4 representative cases of varying difficulty levels for each category to ensure comprehensive coverage of typical scenarios (see Supplementary Table 5 and Supplementary Data 2). Thus, the cases were distinct from one another, and LLMs were expected to behave very differently. Strict privacy protection measures were implemented for all medical records used in this study to ensure patient confidentiality. All personally identifiable information—including, but not limited to, names, addresses, phone numbers, national identification numbers, hospital identifiers, physician identifiers, medical record numbers, hospitalization numbers, and specific dates—was systematically removed through a combination of automated scripts and manual review by trained clinical researchers. Specific dates were replaced with time intervals relative to the admission date. The resulting dataset underwent rigorous validation to confirm that no individual could be re-identified. Expert general practitioners provided detailed annotations in alignment with routine practice, covering differential diagnosis, management, patient education, and recommendations for further examination. General practice specialists subsequently identified key response elements, defined scoring criteria, and mapped each component to the relevant competency dimensions. The outputs from the LLM were evaluated and scored by experienced GPs according to these criteria.

The AI Patient Test Set simulated clinical consultations to efficiently evaluate AI performance. Prompts were designed to elicit natural, conversational responses from patient agents while strictly restricting content to case-provided information (see Supplementary Method 2). Each "patient" only answered questions posed by the "doctor" and did not volunteer additional details. A maximum of $T = 10$ inquiry rounds was set to assess the model's ability to extract key information within limited interactions.

Scoring protocol

The clinical cases consisted of anonymized real-world medical records, which were verified and standardized by expert general practitioners (see Supplementary Table 6). Each case was carefully reviewed and refined to enhance its quality, in accordance with standards for medical training. Before the evaluation, detailed scoring criteria were developed through a Delphi process to ensure a consistent understanding of all evaluation dimensions and rules. The scoring panel was composed of three licensed GPs, referred to as GP-1, GP-2, and GP-3. GP-1 and GP-2 each had over ten years of clinical experience, while GP-3 was a senior GP with fifteen years of experience. All evaluators underwent training on the criteria using a set of randomly selected sample cases to align their interpretations and maintain scoring consistency. During the evaluation, GP-1 and GP-2 independently scored each response. If a discrepancy exceeding 10 points occurred for any individual item, GP-3 conducted arbitration, or the panel convened to reach a consensus (see Supplementary Data 3).

Outcome measures

For the MCQ Test Set, accuracy was calculated based on responses to the corresponding questions. Each multiple-choice question was prompted five times for each LLM with different orders of answer options, and the final prediction for each question was determined by majority voting across the five responses. To conduct a more precise analysis across different competency dimensions, each question was labeled with its corresponding competency indicator. Evaluation was then based on the weighted average accuracy across these indicators. The reported weighted average and standard deviation summarize each model's performance across all questions, reflecting the variance among questions rather than the variance introduced by repeated prompting.

For the Clinical Case Test Set, final scores were assigned by the scoring panel in strict accordance with predefined scoring criteria and the relative weights of each indicator. Each case was evaluated once without repeated sampling. All annotations were independently validated by a separate group of experts. The reported weighted average and standard deviation summarize model performance across all cases, reflecting variability in performance among cases.

For the AI Patient Test Set, experts assessed the competency of LLMs in conducting medical history taking (see Supplementary Table 4). Each "AI patient" agent was evaluated once, and the reported average and standard deviation summarize performance across all agents.

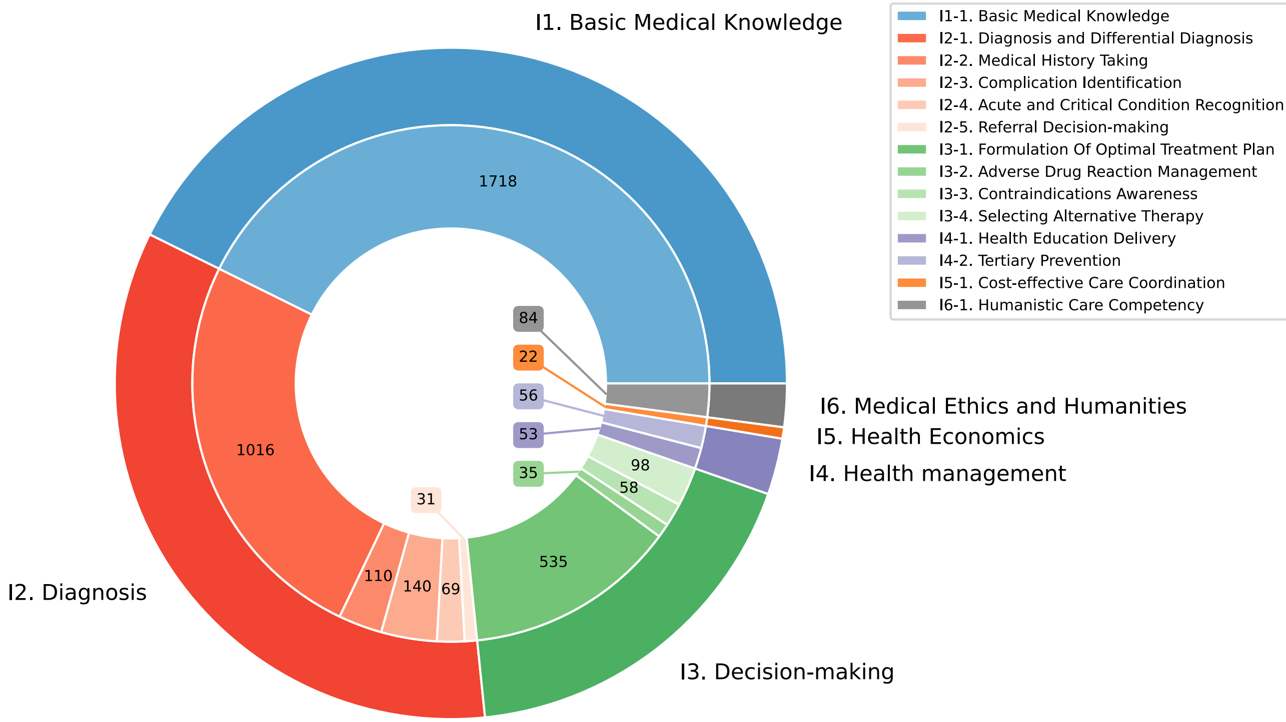

Fig. 6 | The data distribution of the Multiple-choice Question (MCQ) Test Set on different competencies.

Human evaluation

To perform a comparative analysis of the performance of LLMs, we conducted a human evaluation benchmark involving general practitioners (see Supplementary Method 4). Participants were categorized into three groups based on their clinical experience: Group A (more than 10 years of experience, 2 participants), Group B (5 to 10 years of experience, 2 participants), and Group C (less than 5 years of experience, 2 participants). During the evaluation, 200 multiple-choice questions were selected from the MCQ test set through stratified random sampling to ensure coverage of all competency dimensions. The full set of 70 cases from both the Clinical Case Test Set and the AI Patient Test Set was utilized. All physicians completed the evaluation independently in a quiet environment and received standardized instructions that were nearly identical to those provided to the LLMs. The scoring process for human responses strictly followed the same procedures and criteria used for evaluating the LLM outputs (see Section 4.4), ensuring the results are directly comparable.

Statistics & reproducibility

In the MCQ Test Set, the accuracies of the fourteen indicators were assumed to be independent and to follow binomial distributions with varying parameters. Welch's t-test was employed to compare the performance between different LLMs. In the Clinical Case Test Set and AI Patient Test Set, the performance of each LLM was summarized using the mean and standard deviation of the case scores. As the medical records were deliberately selected to include diverse diseases and symptoms, the score distributions significantly deviated from a Gaussian distribution, as confirmed by the Shapiro-Wilk test ($\alpha = 0.05$). Consequently, the Wilcoxon rank-sum test was used for comparisons between LLMs. All statistical analyses were conducted using Python 3.12.4 and SciPy 1.14.1.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The prompts used for LLMs have been included in the manuscript. Both the development set and the test set were made available exclusively for non-commercial research purposes under a data use agreement via our project homepage (https://github.com/AIPrimaryCare). The development set could be accessed in full, including both the evaluation questions and the corresponding scoring rubrics. To preserve evaluation integrity, only the evaluation questions of the test set were released. Access requests should be submitted to the corresponding author, accompanied by a detailed description of the research purpose. The corresponding author and the data-contributing institutions review the request and determine whether access can be granted. Source data are available at the Source Data file. Source data are provided with this paper.

Code availability

The source code for the study is available in the GitHub repository at https://github.com/AIPrimaryCare/gpbench_code. To ensure long-term accessibility and reproducibility, the repository has been archived on Zenodo under the https://doi.org/10.5281/zenodo.18428084. A comprehensive README file is provided with full instructions for reproducing the experiments.

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Author contributions

Z. Li wrote the original draft, performed the writing, review, and editing, and developed the methodology. Y.Y. carried out the data curation, performed the formal analysis and visualization, and wrote the original draft, review, and editing. J. Lang performed the review and editing, carried out the data curation, and conducted the investigation and validation. W.J. conceived the study, developed the methodology, performed the review and editing, carried out the formal analysis, and supervised the work. J. Chen conducted the investigation and validation, developed the methodology, and performed the review, and editing. Y.Z. carried out the data curation, conducted the investigation, and developed the software. D.W. carried out the data curation, developed the software, and performed the visualization. S. Li conducted the investigation. Z. Lin conducted the investigation. X. Li conducted the investigation. Y.T. conducted the investigation. J.Q. conducted the investigation. X. Lu conducted the investigation. H.Y. conducted the investigation. S. Chen conducted the investigation. Y.B. conducted the investigation. X.Z. conducted the investigation. Y. Chen carried out the data curation. L.Y. conceived the study, developed the methodology, provided the resources, supervised the work, and acquired the funding. Data verification: Y.Y., J. Lang, J. Chen, and Y.Z. directly accessed and verified the underlying data.

Competing interests

The authors declare no competing interests.

Additional information

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